J

ITMA 95

SCREENING TEST - JUNIOR LEVEL - 1995

8984

1. Let $S = 1^4 + 2^4 + \cdots + 1995^4$. When S is divided by 10, the remainder is

(A) 6 (B) 4 (C) 2 (D) 0

2. $2^{1/3} + 4^{1/3}$ is a root of

(A) $x^3 + 6x - 6 = 0$ (B) $x^3 - 6x - 6 = 0$ (C) $x^3 + 6x + 6 = 0$ (D) $x^3 + x - 1 = 0$

3. The roots of $x^2 + 3|x| - 3 = 0$ are

(A) $\frac{-3 \pm \sqrt{21}}{2}$ (B) $\frac{\pm 3 + \sqrt{21}}{2}$ (C) $\frac{-3 + \sqrt{21}}{2}, \frac{3 - \sqrt{21}}{2}$ (D) $\frac{3 + \sqrt{21}}{2}, \frac{-3 - \sqrt{21}}{2}$

4. The value of $1.2 + 2.3 + \cdots + 99.100$ is

(A) 300000 (B) 330000 (C) 333000 (D) 333300

5. The smallest positive integer with 24 divisors (where 1 and the number itself are also considered as divisors) is

(A) 420 (B) 240 (C) 360 (D) 480

6. The number of integers $k, 1 \leq k \leq 125$, such that k and 125 have no common divisors other than 1 is

(A) 124 (B) 100 (C) 90 (D) 80

7. The number of integers between 2000 and 8000 which have atleast two digits equal is

(A) 2977 (B) 2342 (C) 2183 (D) 2096

8. If $x^2 + 2x + 5$ is a factor of $x^4 + px^2 + q$, the values of p and q are respectively

(A) –2,5 (B) 5,25 (C) 10,20 (D) 6,25

9. Let $p = 2 \cdot 3 \cdot 5 \cdots 61$, product of all primes less than or equal to 61. The number of primes in the sequence

$$p+2, p+3, p+4, \cdots p+59 \text{ is}$$

(A) 0 (B) 16 (C) 17 (D) 57

10. A student on vacation for d days observed that
(1) it rained 7 times morning or afternoon
(2) when it rained in the afternoon, it was clear in the morning
(3) there were five clear afternoons
(4) there were six clear mornings.
Then d equals

(A) 7 (B) 9 (C) 10 (D) 11

11. AB is the hypotenuse of a right angled $\triangle ABC$. D, E are mid points of BC and CA respectively. $AD = 7$ and $BE = 4$. The length of AB is

(A) 10 (B) $5\sqrt{3}$ (C) $5\sqrt{2}$ (D) $2\sqrt{13}$

12. Two perpendicular chords intersect in a circle. The segments of one chord are 3 and 4; the segments of the other are 6 and 2. The diameter of the circle is

(A) $\sqrt{65}$ (B) $\sqrt{75}$ (C) $\sqrt{61}$ (D) $\sqrt{56}$

13. From a two-digit number N we subtract the number with the digits reversed and find that the result is a non-zero positive perfect cube. Then:

(A) N cannot end in 5
(B) There is no such number
(C) There are exactly 7 such numbers
(D) There are exactly 10 such numbers

14. If the least possible value of $x^2 + px + q$ is zero, q is

(A) $\frac{p^2}{4}$ (B) $\frac{p}{2}$ (C) 0 (D) $\frac{-p}{2}$

15. The remainder when $(x^2 + x - 1)^{100}$ is divided by $x^2 - 1$ is

(A) 1 (B) x (C) $x + 1$ (D) $x - 1$

16. The external length, breadth and height of a closed box are 8 cm, 6 cm and 7 cm respectively. The thickness of the sides is 0.5 cm. The inner surface area is

(A) 200 sq cm (B) 214 sq cm (C) 204 sq cm (D) 194 sq cm

17. The number pair $\{a, b\}$ of integers such that $1 \leq a, b \leq 1800, a.\ b = 1800$, is (when e.g. $\{2, 900\}, \{900, 2\}$ are not considered as distinct pairs) is

(A) 18 (B) 36 (C) 72 (D) 96

18. In a $\triangle ABC, AB = \sqrt{19}, AC = \sqrt{7}, BC = 6. D, E$ are points on BC such that $\triangle ADE$ is equilateral. DE equals

(A) 1 (B) 2 (C) 3 (D) 4

19. An integer x such that $\dfrac{1}{x-2} + \dfrac{1}{x} + \dfrac{1}{x+3} = \dfrac{1}{4}$ is

(A) -1 (B) 1 (C) 3 (D) 4

20. $x \neq 1$. The product $(1+x)(1+x^2)(1+x^4)(1+x^8)(1+x^{16})$ equals

(A) $\dfrac{1-x^{32}}{1-x}$ (B) $\dfrac{1-x^{16}}{1-x}$ (C) $\dfrac{1-x^8}{1-x}$ (D) $\dfrac{1+x^8}{1+x}$

21. If A, B are two sets, $A - B = \{x : x \in A, x \notin B\}$ and $A \triangle B = (A-B) \cup (B-A)$. Let $A = \{1,3,5,7\}, B = \{1,2,3,4,\}, C = \{4,5,6,7\}. A\triangle(B\triangle C)$ equals

(A) $\{2,6\}$ (B) $\{1,3,5\}$ (C) $\{1,2,6\}$ (D) $\emptyset$

22. Let $S = \{4, 8, 12, 16, 36, 32, 108, 64, \cdots\}$. The 18th member in the above sequence is

(A) 65536 (B) 4.3^{17} (C) 2^{11} (D) 2^{19}

23. A number which when divided by 10 leaves a remainder 9, when divided by 9 leaves a remainder 8, by 8 leaves a remainder 7 etc. down to when divided by 9 leaves a remainder 1, is

(A) 59 (B) 419 (C) 2519 (D) 1259

24. $\dfrac{\sqrt{2}}{\sqrt{2}+\sqrt{3}-\sqrt{5}}$ equals

(A) $\dfrac{3+\sqrt{6}+\sqrt{15}}{6}$ (B) $\dfrac{\sqrt{6}-2+\sqrt{10}}{6}$ (C) $\dfrac{2+\sqrt{6}+\sqrt{10}}{10}$ (D) $\dfrac{2+\sqrt{6}-\sqrt{10}}{6}$

25. If $\left(a+\dfrac{1}{a}\right)^2 = 3, a^3+\dfrac{1}{a^3}$ equals

(A) $\dfrac{10\sqrt{3}}{3}$ (B) $3\sqrt{3}$ (C) $7\sqrt{7}$ (D) 0

26. The number of scalene Δs having all sides of integral lengths and perimeter less than 13 is

(A) 18 (B) 2 (C) 3 (D) 4

27. If $(1-x)(1-2x)\cdots(1-10x) = a_0 + a_1x + \cdots a_{10}x^{10}, a_2 + a_3 + a_4 + \cdots + a_{10}$ is

(A) 50 (B) -10 (C) -54 (D) 54

28. $xy(x+y) + yz(y+z) + zx(z+x) + 2xyz$ equals

(A) $(x+y)(y+z)(z+x)$
(B) $(x+y)(y+z)(z+x) - xyz$
(C) $x^2(y+z) + y^2(z+x) + z^2(x+y)$
(D) none of these

29. If $\dfrac{29}{17} = 1 + \dfrac{1}{1+\dfrac{1}{x+\frac{x}{y}}}$ where x, y are positive integers, then x, y are respectively

(A) 5,2 (B) 2,5 (C) 2,3 (D) 3,2

30. a, b, c are distinct real numbers. $\dfrac{(a-b)(b-c)(c-a)}{(a-b)^3+(b-c)^3+(c-a)^3}$ equals

(A) 3 (B) $\dfrac{1}{3}$ (C) 1 (D) none of these.